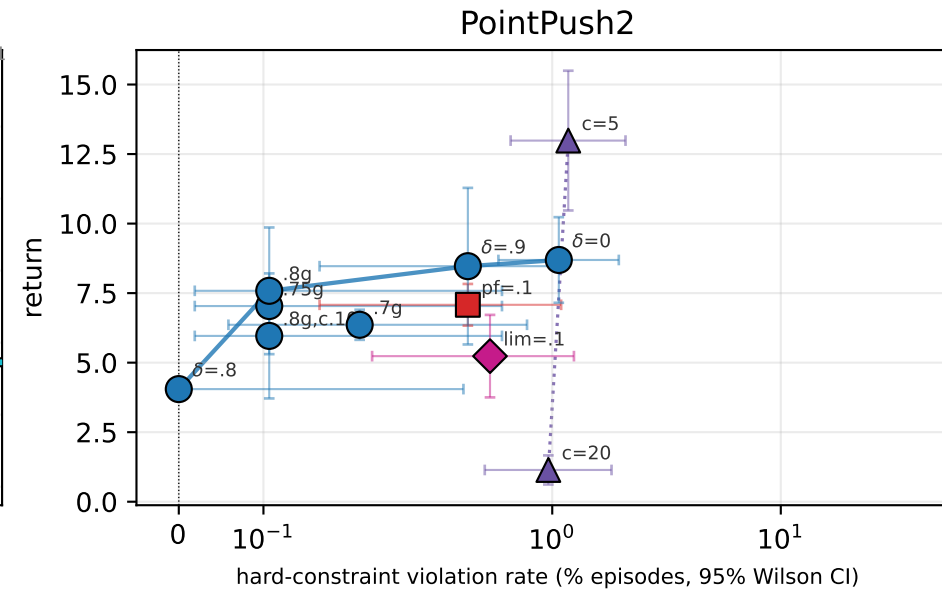


Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost > 25/episode)



- SafeOT-typed (ours)
- FDPI (constraint-based)

- SafeOT-typed (ours)
- FDPI (constraint-based)
- SAC-Lagrangian (penalty-based)
- SAC + fixed penalty (penalty-based)

- SDAC (multi-constraint SOTA)
- SRCPPO (multi-constraint SOTA)

 SDAC (multi-constraint SOTA)  SAC-Pen coefficient sweep
 SRCPO (multi-constraint SOTA)  FDPI p_f sweep
 typed Pareto frontier (soft ≤ 25)  SAC-Lag cost-limit sweep

hard-constraint violation rate (% episodes, 95% Wilson CI)

- • • • • SAC-Pen coefficient sweep
- FDPI p_f sweep
- SAC-Lag cost-limit sweep

hard-constraint violation rate (% episodes, 95% Wilson CI)

Constraint violation rate (% episodes)